

MSE 413 Final Project Proposal

Emotion Analysis Classification Based on Facial Coordinates

Name: Tom Dang
Student ID: 301548582

Name: Sam Drubetsky
Student ID: 301452017

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Introduction

This project aims to classify facial expressions based solely on mouth shape. We will use computer vision-based facial landmark detection tools to quantify facial expressions into sets of coordinate data. Then, we can apply machine learning techniques on coordinate data to create several models and even plot the relations of certain manually defined features (such as mouth angle, eye width, eye tilt, etc.).

Dataset

- **Dataset Name:** Human Face Emotions
- **Source:** <https://www.kaggle.com/datasets/sanidhyak/human-face-emotions>
- This dataset consists of labeled images of faces displaying different emotions.

Facial Landmark Detection

To extract mouth-related features, we will use the computer vision based:

- **Dlib's 68-point facial landmark model**
 - Provides key points around the mouth and other facial features
 - OpenCV-based implementation
 - Allows extraction of numerical data from facial images

This method effectively transforms facial images into an array of coordinate points indicating the position and orientation of facial features. Thereby allowing us to create a model of our choosing, ~ KNN models, Regression models, Random forest models, etc.

Feature Extraction

Once mouth landmarks are detected, we will compute the geometric features that we think are important to then contrast against what coordinates our model prioritized. Features include:

- **Mouth width:** Distance between the two corners of the mouth.
 - Formula: `Width = distance(landmark[48], landmark[54])`
- **Mouth height:** Distance between the upper and lower lips.
 - Formula: `Height = distance(landmark[51], landmark[57])`
- **Mouth curvature:** An Angle or arc formed by the upper lip.
 - It can be approximated using polynomial fitting or Bezier curves.

These features will serve as custom input variables for machine learning models and will be used to compare how efficiently a model can determine expression using preconceived features as opposed to a sporadic set of coordinates.

Machine Learning Models

K-Nearest Neighbors (KNN) and Polynomial Regression or Support Vector Regression (SVR)

- Works well for visualizations as it can be plotted in 2D space.
- It may face the curse of dimensionality with too many features. It may prove to be very useful to compare custom features

Decision Trees & Random Forests

- It can handle complex feature interactions.
- Likely to be the most accurate but less interpretable with simple plots.
- Extracts patterns from the dataset rather than requiring manual feature engineering.

Visualization & Analysis

- Several 2D plots will be created to show feature distributions and classifications.
- Additional visualizations can be introduced after running high-dimensional data.
- Comparisons will be made between different algorithms based on accuracy and interpretability.

Conclusion

This project aims to use machine learning to analyze and classify facial expressions using mouth shape as the primary feature. By using several models, we hope to identify which models are most effective for this kind of data analysis and also compare a model's conclusion regarding which features are most important as opposed to our initial preconceptions.